

Supply Chain Logistics Manager - Functional Requirements v1

Confidential - Internal Use Only

Version 1.0

1. Platform Overview

Enterprise logistics platform for managing shipments across FedEx, UPS, DHL, and regional carriers.

Serves 500+ enterprise customers with 2 million+ monthly shipments across 45 countries.

Real-time tracking, warehouse management, route optimization, and customs automation.

2. Shipment Management

REQ-LOG-001: Create shipment with origin, destination, weight, dimensions, and service level.

REQ-LOG-002: Multi-piece shipment support with individual tracking per piece.

REQ-LOG-003: Shipment status lifecycle: Created, Picked Up, In Transit, Out for Delivery, Delivered, Exception.

REQ-LOG-004: Proof of delivery capture with photo, signature, and GPS coordinates.

REQ-LOG-005: Shipment cloning for recurring routes with saved templates.

3. Real-Time Tracking

REQ-LOG-006: Live GPS tracking displayed on interactive Google Maps with vehicle icons.

REQ-LOG-007: ETA calculation updated every 15 minutes based on traffic and weather.

REQ-LOG-008: Geofence alerts when shipment enters or exits defined zones.

REQ-LOG-009: Tracking page shareable via public URL with customer branding.

REQ-LOG-010: Push notifications for status changes to shipper and consignee.

4. Carrier Integration

REQ-LOG-011: API integration with FedEx, UPS, DHL, USPS, and Canada Post.

REQ-LOG-012: Real-time rate comparison across all integrated carriers.

REQ-LOG-013: Automatic label generation in PDF and ZPL (thermal printer) format.

REQ-LOG-014: Carrier performance scorecards (on-time %, damage rate, cost per kg).

REQ-LOG-015: Carrier SLA monitoring with breach alerting.

5. Route Optimization

REQ-LOG-016: AI-powered route optimization minimizing cost, time, or carbon footprint.

REQ-LOG-017: Multi-stop route planning with time window constraints per stop.

REQ-LOG-018: Dynamic re-routing when delays or disruptions are detected.

REQ-LOG-019: Route optimization considers vehicle capacity, driver hours, and fuel costs.